

Bhavith V

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Mangaluru, Karnataka, India

EDUCATION

Canara Engineering College <i>Bachelor of Engineering in Computer Science and Engineering</i>	Mangaluru, Karnataka, India 2023 – Present
St. Aloysius PU College <i>Pre-University Course (PCME) — 91.33%</i>	Mangaluru, Karnataka, India 2021 – 2023

TECHNICAL SKILLS

Programming: Python, C, Bash
Web Technologies: HTML5, CSS3, MERN Stack
Databases: MongoDB, Supabase
Cybersecurity: Linux (CLI), API Security Basics, Secure Coding Practices
Developer Tools: Git, GitHub, VS Code, Docker (Containerization & Environment Simulation)
Cloud Platforms: Vercel, Render, Streamlit Cloud

CYBERSECURITY SKILLS

- Network Fundamentals: TCP/IP, DNS, HTTP communication basics
- Packet analysis and traffic inspection using Wireshark
- Linux system monitoring and log inspection
- Basic vulnerability analysis and secure backend design
- Understanding of access control, API security, and secure development practices

PROJECTS

Posture AI: Intelligent Posture Detection and Feedback Tool <i>MERN, TF.js, WebRTC</i>	2025 – 2026
• Developed a web-based application utilizing TensorFlow.js and MoveNet pose estimation to provide real-time posture analysis and form correction across six exercise types. • Engineered a trainer dashboard capable of monitoring 20+ concurrent trainees remotely, integrating WebRTC for peer-to-peer video streaming and Socket.IO for real-time data transfer. • Implemented custom algorithms to calculate joint angles and detect movement phases, achieving 90–94% accuracy at 15–20 frames per second using standard webcams. • Integrated voice-guided feedback, Jitsi Meet, and automated PDF report generation to deliver professional-level fitness guidance without expensive hardware.	
SilentSOS-Med – Real-Time Emergency Detection <i>React, Node.js, TF.js, 1D-CNN/Bi-LSTM</i>	Ongoing
• Designed a wearable-free mobile architecture that autonomously detects medical emergencies by mathematically isolating dynamic acceleration from smartphone inertial sensors. • Deployed an on-device Machine Learning pipeline using a 1D-CNN/Bi-LSTM architecture and a continuous 100-frame sliding window for offline-capable, zero-latency inference. • Built a debounce-locked alert cascade that triggers at an 85% confidence threshold, natively dispatching SMS with real-time GPS coordinates to emergency contacts. • Routed telemetry through a Node.js backend to trigger instantaneous, WebSocket-driven notifications on a scalable React-based Caregiver Web Dashboard.	
Autonomous Self-Healing Infrastructure using DRL <i>Python, Docker, DRL (PPO/DQN)</i>	Ongoing
• Proposed an AI-driven autonomous system for educational institutions to proactively monitor server telemetry (CPU, memory, disk I/O) and prevent downtime during traffic bursts. • Implemented a Deep Reinforcement Learning agent (PPO/DQN) within simulated Docker environments to execute budget-aware self-healing actions such as resource scaling and service restarts. • Incorporated context-awareness to balance service availability and cost efficiency by considering academic calendar events and distinguishing between valid traffic and anomalies. • Developed real-time visualization dashboards to display server health metrics, predicted issues, and provide explainable AI insights behind automated mitigation decisions.	

- Developed an intelligent web-based chatbot integrating Google's Gemini 2.0 Flash API with a JSON database to streamline access to academic schedules and campus events.
- Optimized API token management and resolved complex state synchronization issues to successfully handle 150+ daily personalized queries within free-tier constraints.
- Deployed on Streamlit Cloud using GitHub CI/CD, reducing information retrieval time from 15+ minutes to under 2 seconds while maintaining 24/7 availability.

ACHIEVEMENTS AND EXPERIENCE

- **1st Prize, Mini Project Expo (Dec 2025):** Led the development of 'PostureAI', an intelligent real-time monitoring system, securing first place in the Computer Science and Engineering Department at Canara Engineering College.
- **Open Source Contributor (Oct 2025):** Participated in Hacktoberfest 2025, collaborating on GitHub repositories, reviewing code contributions, and maintaining secure version control workflows.
- **Security Mindset in Development:** Consistently applied access control, real-time event monitoring, and API security across full-stack applications.

CERTIFICATIONS

- **Python from Zero-to-Hero** – Udemy (Aug 2025)
- **C Programming Course** – Infosys Springboard (May 2024)
- **Foundations of Cybersecurity** – Google & Coursera (May 2024)